

Hardware Project No. 3: Passive High-Pass Filter

This time, we try to kick it really old school (olde school?) by building a passive high-pass filter. We will be somewhat constrained by inductor availability but the goal will be to build a third-order, high-pass Butterworth filter. One additional constraint: so you can feel the true pain of dealing with inductors, the single inductor you need to use shall be built of a series or parallel (your choice) combination of two inductors (why not just do a fourth-order filter, which would require two capacitors and two inductors? Because you'd likely be stuck trying to make one of those inductors out of multiple inductors, and I don't hate you that much).

Refer to the attached sheets, reprinted from *Introduction to the Theory and Design of Active Filters*, by Huelsman and Allen (1980) for component values for the circuit you will build. Note that these tables are for building *low-pass* filters; you will need to apply the low-pass-to-high-pass transformation to build high-pass. Recall from a previous discussion, that changes inductors to capacitors with $C=1/L$ and capacitors to inductors with $L=1/C$.

Apply the frequency and impedance denormalizations to get component values you can actually use (i.e., you know pretty much what inductors you have to work with, so go backwards from there). The tables give you values for a 1 rad/sec cutoff frequency; a good choice for measurements we would like to make, such as EMG, would be something like 20 Hz; however, that would lead to very few measurements in the transition and stop bands of the filter. Since it would be difficult to make so many very low-frequency measurements, pick something that is easily achievable with available inductors, such as 100 or 500 Hz. You should note this in your report. Use an impedance denormalization that gets you reasonable capacitors (however, your impedance denormalization may have been forced upon you by the inductors you have). Build the required resistor by whatever means necessary.

Show me your design process, i.e.:

- The table showed certain values
- To get the proper corner frequency, how did you apply a frequency denormalization?
- To get your inductor value, what impedance denormalization did you apply?
- What component values did that give you?
- What components were you able to find? Did that make you go back and redo the impedance denormalization?

Bear in mind, if you end up with a relatively low load resistance (say, much less than 500 Ω) you may be setting yourself up for trouble, as you may be placing a load on the measuring devices: the output impedance of your function generator and input impedance of your oscilloscope may be interesting to know here.

When using multiple inductors, things that change how they interact: distance, orientation relative to each other (parallel or perpendicular), use of magnetic shielding. To measure

your compound “inductor” made from multiple inductors, you will not want to measure them individually and add the values, but rather measure them in their configuration on the board.

DON'T FORGET!!!! Those inductors will have a resistance (you can measure it with a meter). Since all these filters basically are setting up a resonance between the components, this resistance that you cannot get rid of that is in the middle of everything will mess that up to some extent. This is definitely worth putting in a SPICE model, and may explain a lot of whatever deviation your filter has from expectations.

As for the testing, you will need to provide the frequency response of the filter; be sure to measure lots of points around the corner frequency, and make some measurements to show that the stopband falls off at the rate it should depending on the filter order.

Write a report as per the previous hardware assignment.

Due Date

April 2, 2026.